
Inferential Seamless Designs for Treatment Selection during a Phase II/III Study in Oncology

STA495 Master Thesis Proposal

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**University of
Zurich** UZH



Bristol Myers Squibb

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Abbreviations

BMS Bristol Myers Squibb

CI Confidence Interval

GBDS Global Biostatistics and Data Science

HR Hazard Ratio

MAMS Multi-Arm Multi-Stage

OS Overall Survival

PFS Progression-Free Survival

R R Programming Language

SoC Standard of Care

UZH University of Zurich

1 Abstract

The development of new treatments in oncology faces challenges due to the need for evaluating multiple drug combinations, schedules, and dose levels. This proposal investigates the application of Multi-Arm Multi-Stage (MAMS) trials, specifically focusing on the "Drop-the-losers" design introduced by Professor Wason in 2017, which facilitates treatment selection through adaptive randomization. The goal is to assess the efficiency and statistical robustness of various designs, emphasizing reduced sample sizes, increased power, and shorter study durations.

The "Drop-the-losers" approach involves interim analyses where ineffective treatments are dropped, optimizing resource allocation and maintaining statistical power. The proposal outlines the derivation of covariances for continuous, binary, and survival outcomes, and the implementation of the design for these outcomes using R. A comprehensive simulation study will evaluate the designs, focusing on Type I and II errors, sample size, and study duration.

This research will utilize simulated data to ensure confidentiality, with detailed simulation protocols developed for reproducibility. The study aims to identify the most efficient design for Phase II/III oncology trials, potentially extending to p-value combination. The findings will contribute to optimizing clinical trial designs, enhancing the development of effective treatments in oncology.

Keywords

MAMS, Phase 2/3 study, Treatment selection, Dose selection, Drop-the-losers, Oncology

2 Introduction

The introduction of new assets has brought forth challenges in clinical development, including the exploration of various drug combinations, schedules, and dose levels. Utilizing treatment selection designs, which compare multiple experimental treatments concurrently against a standard of care (SoC) control group, stands out as a promising approach to enhance the clinical development strategy.

A Multi-Arm Multi-Stage (MAMS) trial is an efficient approach in clinical research that tests several treatments against a control group within the same study. This design allows for interim analyses at various stages, enabling researchers to adapt the trial based on early results. Poorly performing treatments can be dropped, focusing resources on more promising ones. MAMS trials, with their inferential seamless design advantage, are particularly useful for quickly identifying effective treatments without starting a new trial. This makes them ideal for areas with multiple potential therapies, such as oncology, or public health emergencies, as they allow for the reuse of data. This design not only saves time and resources but also maintains high statistical power despite its complexity (Millen and Yap, 2020).

In response, Professor Wason introduced a novel design in 2017 known as "Drop-the-losers," a specific iteration of adaptive randomization design (Wason et al., 2017). This design features interim analyses where less effective experimental treatments are

assigned a 0% randomization chance, allowing only the more promising treatments to continue with equitable randomization probabilities. This streamlined process ensures the study design's robust performance, including stringent Type I error control, as demonstrated in scenarios with only normal endpoints in the publication. The firm control over Type I errors renders this design particularly suitable for Phase II/III studies, enabling an inferential seamless approach.

Another feature of this design is that, thanks to predefined numbers of selected treatment arms after each interim analysis, the overall sample size is fixed. This characteristic, though counter-intuitive for an adaptive design, is very convenient when discussing the study's timeline and resources.

The “Drop-the-losers” design, denoted as 4:2:1, involves starting with four experimental treatments in Stage 1, narrowing down to two in Stage 2, and concluding with a single treatment for the final analysis. The design specifies the number of participants per treatment arm and stage, resulting in a predetermined total sample size:

$$N = \sum_{i=0}^4 n_{1i} + \sum_{i \in \{0,1,3\}} n_{2i} + \sum_{i \in \{0,3\}} n_{3i}$$

- N : Total sample size for the study.
- n_{ji} : Sample size for treatment arm i at stage j .
 - j : Stage of the trial (1, 2, 3).
 - i : Treatment arm (0: SoC, 1: Arm 1, 2: Arm 2, 3: Arm 3, 4: Arm 4).
 - Each arm can involve the same or different patients at each stage of the trial. For example, the SoC group in the first interim analysis (IA1) may include different patients than in the second interim analysis (IA2).

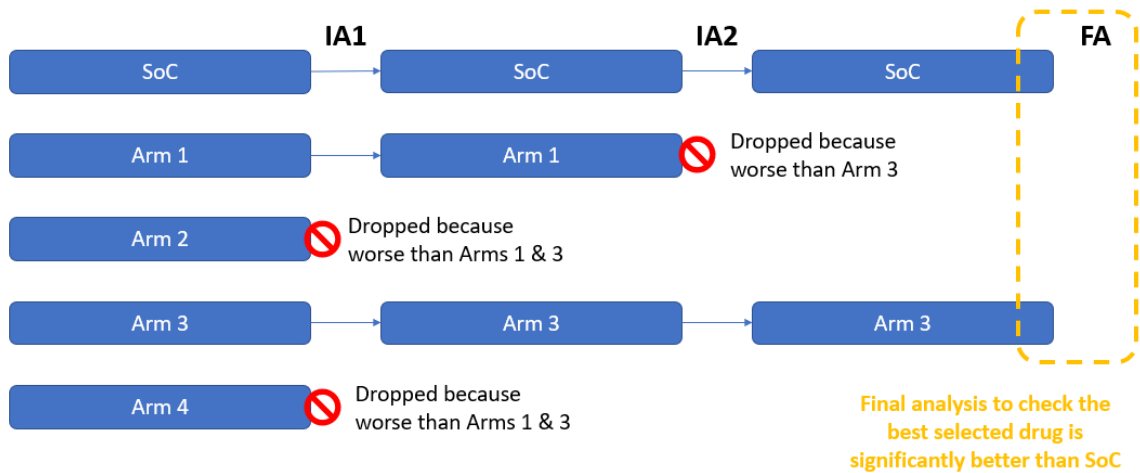


Figure 1: Illustration of a Drop-the-losers design (Yi, 2022)

To calculate the test statistics in a multi-arm clinical trial, you first need to define the null hypothesis H_0 for each treatment i as $H_{0i} : \theta_i \leq \theta_0$, where θ_0 is the effect of the control treatment. The known variance test statistic for treatment i at stage j is calculated using the formula

$$Z_{i(j)} = \frac{\sum_{k=1}^{n_j} X_{ik}/n_j - \sum_{k=1}^{n_j} X_{0k}/n_j}{\sqrt{2\sigma^2/n_j}},$$

where X_{ik} and X_{0k} are the outcomes for the treatment and control groups respectively. This statistic follows a normal distribution with mean $\theta_i\sqrt{n_j/2\sigma^2}$ and variance 1.

To compare test statistics at each stage, the covariance between them is calculated using the formula¹(Wason et al., 2017):

$$\text{Cov}(Z_{i(j)}, Z_{m(l)}) = \begin{cases} \sqrt{\frac{\min(n_j, n_l)}{\max(n_j, n_l)}} & \text{if } i = m; \\ \sqrt{\frac{1}{2} \frac{\min(n_j, n_l)}{\max(n_j, n_l)}} & \text{if } i \neq m \end{cases}$$

At each interim analysis, a fixed number of treatments with the smallest test statistics are dropped, ensuring that only the most promising treatments continue. The control treatment remains throughout the trial, and if the final test statistic for the remaining treatment exceeds a predefined threshold, the treatment is recommended and the null hypothesis is rejected. This method ensures control over the family-wise type I error rate (FWER), maintaining it below a specified significance level.

To control the FWER, the design ensures that the probability of rejecting at least one true null hypothesis among all treatments is kept below the pre-specified significance level α . This is achieved by setting appropriate critical values for the test statistics at each stage of the trial. By using a pre-specified schedule to drop a fixed number of treatments at each interim analysis, the design controls the FWER by ensuring that the probability of making one or more type I errors does not exceed the nominal level.

A more commonly used approach is the p-value combination (Bauer and Kieser, 1999; Jaki and Magirr, 2013). However, this approach is usually described as a single efficacy analysis. Extending the p-value combination approach based on the Group Sequential Design method and related alpha-spending functions may provide another good option to deal with treatment selection during a Phase II/III study.

Illustration of combination test:

The concept of p-value combination involves methods for combining p-values from different statistical tests to assess the overall evidence against a null hypothesis. One commonly used method is Fisher's combination test, which combines independent p-values $p_1, p_2, \dots, p_k$ using the formula:

$$X = -2 \sum_{i=1}^k \ln(p_i)$$

¹The formula was copied from the paper. For the case of unequal arm (lower formula), the formula should be without the square root

The test statistic X follows a chi-squared distribution with $2k$ degrees of freedom under the null hypothesis. Another approach is Stouffer's method, which combines p-values by converting them to z-scores, summing them, and then standardizing the result:

$$Z = \frac{\sum_{i=1}^k \Phi^{-1}(p_i)}{\sqrt{k}}$$

where Φ^{-1} is the inverse standard normal cumulative distribution function. These methods help in aggregating the evidence from multiple tests to make a more robust decision regarding the null hypothesis (Bauer and Kieser, 1999).

Fisher's Method: (Bauer and Kieser, 1999)

1. Calculate the test statistic using the combined p-values:

$$X = -2 \sum_{i=1}^k \ln(p_i)$$

2. Determine the adjusted p-value by finding the cumulative distribution function (CDF) of the chi-squared distribution with $2k$ degrees of freedom:

$$p_{\text{adj}} = 1 - F_{\chi_{2k}^2}(X)$$

where $F_{\chi_{2k}^2}$ is the CDF of the chi-squared distribution with $2k$ degrees of freedom.

Stouffer's Method: (Bauer and Kieser, 1999)

We will particularly focus on designs that use Stouffer's Method.

1. Calculate the test statistic using the combined z-scores:

$$Z = \frac{\sum_{i=1}^k \Phi^{-1}(p_i)}{\sqrt{k}}$$

2. Determine the adjusted p-value by finding the cumulative distribution function (CDF) of the standard normal distribution:

$$p_{\text{adj}} = 1 - \Phi(Z)$$

where Φ is the CDF of the standard normal distribution.

Goal

To accurately evaluate the benefits and limitations of various statistical designs for Multi-Arm Multi-Stage (MAMS) clinical trials, we will particularly focus on designs that exclude the 'Drop-the-Losers' approach, with the p-value combination and Bayesian adaptive randomization being optional if enough time is available. To find the best design, this analysis will focus on designs optimized for smaller total sample sizes, increased statistical power, and reduced study duration. The objective is to identify the

most efficient statistical design for Phase 3 studies, specifically aimed at determining the optimal dose or drug combination for treatment selection.

Regarding the level of significance, it is challenging to expect a significant difference between experimental drugs, as all are anticipated to outperform the standard of care. The study is typically not powered to detect significant differences between experimental arms, so a common practice is to select the drug with the highest efficacy, even if the difference is small. We will explore how and when to recommend the number of experimental drugs to advance to Stage 2, whether predefined or based on interim analysis results. For example, if two experimental drugs show similar efficacy, should we recommend both for Stage 2? Additionally, we will consider how to control the Type I error in such situations.

Expected Intermediate Steps

This section provides a high-level overview of the expected intermediate steps, while Table 2 contains more detailed information.

1. Derivation of the covariances for continuous, binary, and survival outcomes. This will involve detailed mathematical formulations for each type of outcome.
2. Implementation of the Drop-the-Loser approach for continuous, binary, and survival outcomes. For continuous outcomes, existing R code by the author Wason will be utilized.
3. Conduct a simulation study to assess the designs' operating characteristics. This will include the setup of a simulation protocol that takes into account time and other factors to ensure accurate calculations of Type I and II errors, sample size, study duration, etc. (These characteristics will be defined more precisely in the protocol).
4. Publication, ensuring the Master's thesis report complies with the University of Zurich's guidelines. This report will then be condensed into a format suitable for publication. However, the final publication is not part of the Master's thesis (this can also occur at a later date).

July					August				September				October				November				December								
CW 27	CW 28	CW 29	CW 30	CW 31	CW 32	CW 33	CW 34	CW 35	CW 36	CW 37	CW 38	CW 39	CW 40	CW 41	CW 42	CW 43	CW 44	CW 45	CW 46	CW 47	CW 48	CW 49	CW 50	CW 51	CW 52				
Start BMS: Onboarding / Literature Review / Training / Finalize Goals and Proposal		Implementation duplicate paper's result from Wason et al		Implement Drop the Loser to Binary and Survival Endpoints		Holidays	Writing Simulation Protocol			Reserve week to mitigate any potential delays		Holidays							Reserve week to mitigate any potential delays						Holidays				
					Prepare Presentation for Intern Symposium		Setup Simulation	Run 1st batch of simulations vs parallel GSDs			Implement competing MAMS		Implement competing MAMS	Run 2nd batch of simulations vs competing methods															
									Include Futility analysis					Writing Paper							Finalize Paper								
														Preparing Mid Term Presentation										Review Paper		Preparing Presentation			
															Mid-Term Present.	Adjust the Drop the Loser Design		Numerical application								Final Presentation BMS / UZH			

Figure 2: Monthly Schedule

Simulation Study

Simulation studies are integral to evaluating the performance of various statistical methods under controlled conditions that mimic real-world complexities. In this proposal, the simulation study will be meticulously designed to ensure it captures the nuanced behaviors and interactions inherent in clinical trials. Key aspects to consider include clearly defining the study's objectives and hypotheses, ensuring the simulation framework aligns with these goals. Generating realistic datasets is crucial; this involves specifying appropriate distributions, sample sizes, and parameter values based on either theoretical considerations or real-world data. The number of simulation runs should be sufficient to provide robust statistical power and precision, often guided by criteria such as bias, mean square error, and coverage probabilities. Moreover, it is essential to establish procedures for data storage and analysis, ensuring reproducibility and transparency of results.

As part of this work, a comprehensive simulation protocol will be developed, detailing each step of the simulation process, from data generation to analysis, to ensure that the study is conducted systematically and its findings are reliable and replicable. In Figure 3, the individual steps that must be included in the protocol are listed. In reference to point 8 in the figure, the sample size and power will be evaluated. Additionally, Bristol Myers Squibb (BMS) provides access to cloud computing resources, offering increased computational power to handle the extensive simulations required for this study (Burton et al., 2006; Morris, White, and Crowther, 2019).

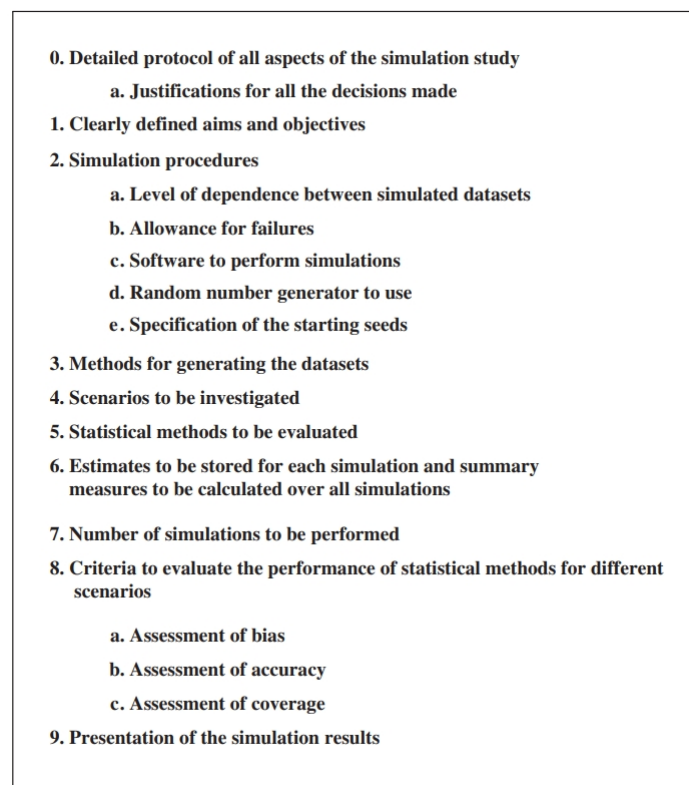
- 
- 0. Detailed protocol of all aspects of the simulation study**
 - a. Justifications for all the decisions made**
 - 1. Clearly defined aims and objectives**
 - 2. Simulation procedures**
 - a. Level of dependence between simulated datasets**
 - b. Allowance for failures**
 - c. Software to perform simulations**
 - d. Random number generator to use**
 - e. Specification of the starting seeds**
 - 3. Methods for generating the datasets**
 - 4. Scenarios to be investigated**
 - 5. Statistical methods to be evaluated**
 - 6. Estimates to be stored for each simulation and summary measures to be calculated over all simulations**
 - 7. Number of simulations to be performed**
 - 8. Criteria to evaluate the performance of statistical methods for different scenarios**
 - a. Assessment of bias**
 - b. Assessment of accuracy**
 - c. Assessment of coverage**
 - 9. Presentation of the simulation results**

Figure 3: Important considerations when designing any simulation study (Burton et al., 2006).

Data

This study primarily uses simulated data due to confidentiality reasons, ensuring the protection of sensitive information. The simulated data includes Progression-Free Survival (PFS) and Overall Survival (OS) metrics, generated using exponential distributions, with OS imputed when occurring before PFS. Additionally, no more metadata is available.

Simulated data and clinical data already published are used as internal data are very likely not to be shared with the university. Therefore, simulated data is used for the publicly accessible version.

The table 1 provides an overview of the data.

Column Name	Description	Details
USUBJID	Unique patient ID. Here it is a dummy one, but it is usually based on the study number and the patient ID (site number + randomization number).	Range: 1 - 650
STAGE	Study part (Stage 1 for treatment selection, stage 2 for final comparison between selected drug and control arm).	Levels: 1, 2
ARM	Treatment group. 1 for control, 2-3-4 for experimental drugs.	Levels: 1, 2, 3, 4
PARAMCD	Parameter Code: Usual variable to handle multiple endpoints in the same dataset.	Levels: OS, PFS
AVAL	Analysis Value: this is the value of interest (to be explained by covariates and/or treatment).	Range: 0.03 - 411.3
CNSR	Censoring flag. 0 when the event occurred. 1 when the event is censored (at the date provided by AVAL).	Levels: 0, 1

Table 1: Details and Overview of the Data

More details about the dataset are applicable in the table 2:

Table 2: Summary Statistics of Simulated Data

Column Name	Statistics
OS	650
PFS	650
STAGE	
1	950
2	350
ARM	
1	512
2	490

Continued on next page

Table 2 – continued from previous page

Column Name	Statistics
3	152
4	146
AVAL	
Min	0.030
1st Quartile	9.928
Median	24.050
Mean	35.806
3rd Quartile	47.750
Max	411.300
CNSR	
0	1042
1	258

Software

The statistical analyses were performed using the open-source software R. The code is stored in a Github repository.

To clone the repository, use the following URL: Github Repository

Dynamic Reporting and Reproducibility

Dynamic reporting tools, such as LaTeX and R Markdown, will be employed to ensure the documentation is dynamically generated for ease of updates. This strategy not only improves document consistency but also aids in the research's reproducibility. All analytical and documentation codes will be provided, enhancing transparency and facilitating future research efforts.

No additional report will be prepared for Bristol Myers Squibb. Should the report be published, Manuel Pfister will serve as the first author. The report primarily features results from simulated data, but a section on Bristol Myers Squibb's clinical data can be included if necessary. Confidential clinical data can be included separately if needed. Additional authors will be considered equal contributors.

The structure of the Master's Thesis Report is designed to align with the guidelines of the University of Zurich and the principles outlined in the paper "Ten Simple Rules for Structuring Papers (Mensh and Kording, 2017)."

Expected deliveries

- Full simulation study to compare different MAMS designs
- R programs to easily implement a MAMS design (with Frequent Statistics, create an R-package is not necessary)
- Final presentation to GBDS department and internship report (we can rely on presentation and report required for Master thesis)

3 Administrative Details

Important Dates

Start Date: 01 July 2024 (for 6 months)

Mid-Term Presentation: The presentation will be held in a hybrid format (will also be available to join via Microsoft Teams) on October 8, 2024, at 10:00 AM at the EPDI Institute at the University of Zurich. The duration will be 45 minutes. The slides will be made available to PD Dr. Malgorzata Roos three weeks prior to the presentation.

Final Examination and Presentation Schedule: The final examination at the University of Zurich is scheduled for the end of December 2024. The presentation will be held in a hybrid format. Additionally, a final presentation is planned to be delivered at BMS in the same month, which will serve as a preparatory rehearsal for the Master's thesis defense.

Regular Meeting Schedule: Initially set every week, the frequency of meetings will be adjusted throughout the Master's thesis. Weekly meetings are particularly recommended during the initial month(s) and the final stage, to assist with the drafting of the Master's thesis manuscript.

Supervision

Bristol Myers Squibb: Dr. Pierre Colin (pierre.colin@bms.com)

University of Zurich: PD Dr. Malgorzata Roos (malgorzata.roos@uzh.ch)

Grading

The master's thesis will be graded by both Dr. Pierre Colin and PD Dr. Malgorzata Roos.

Affiliation

6-month contract with Bristol Myers Squibb, Rte de Perreux 1, 2017 Boudry.

Manuel Pfister is enrolled as a Master's student at the University of Zurich at the same time.

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Adherence to Program Guidelines

This thesis adheres to the guidelines provided in the Master in Biostatistics program's information sheet, which we fully endorse.

Specific rules apply at Bristol Myers Squibb, which are set out in the employment contract.

Financial Compensation

Supervision by the University of Zurich is provided free of charge, with no financial compensation required, provided that the supervising faculty member from the University of Zurich is acknowledged in the resulting publications.

Manuel Pfister

Date / Location: _____ **Signature:** _____

Dr. Pierre Colin

Date / Location: _____ **Signature:** _____

PD Dr. Malgorzata Roos

Date / Location: _____ **Signature:** _____

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